Ethanol Blending in Gasoline – Malaysia

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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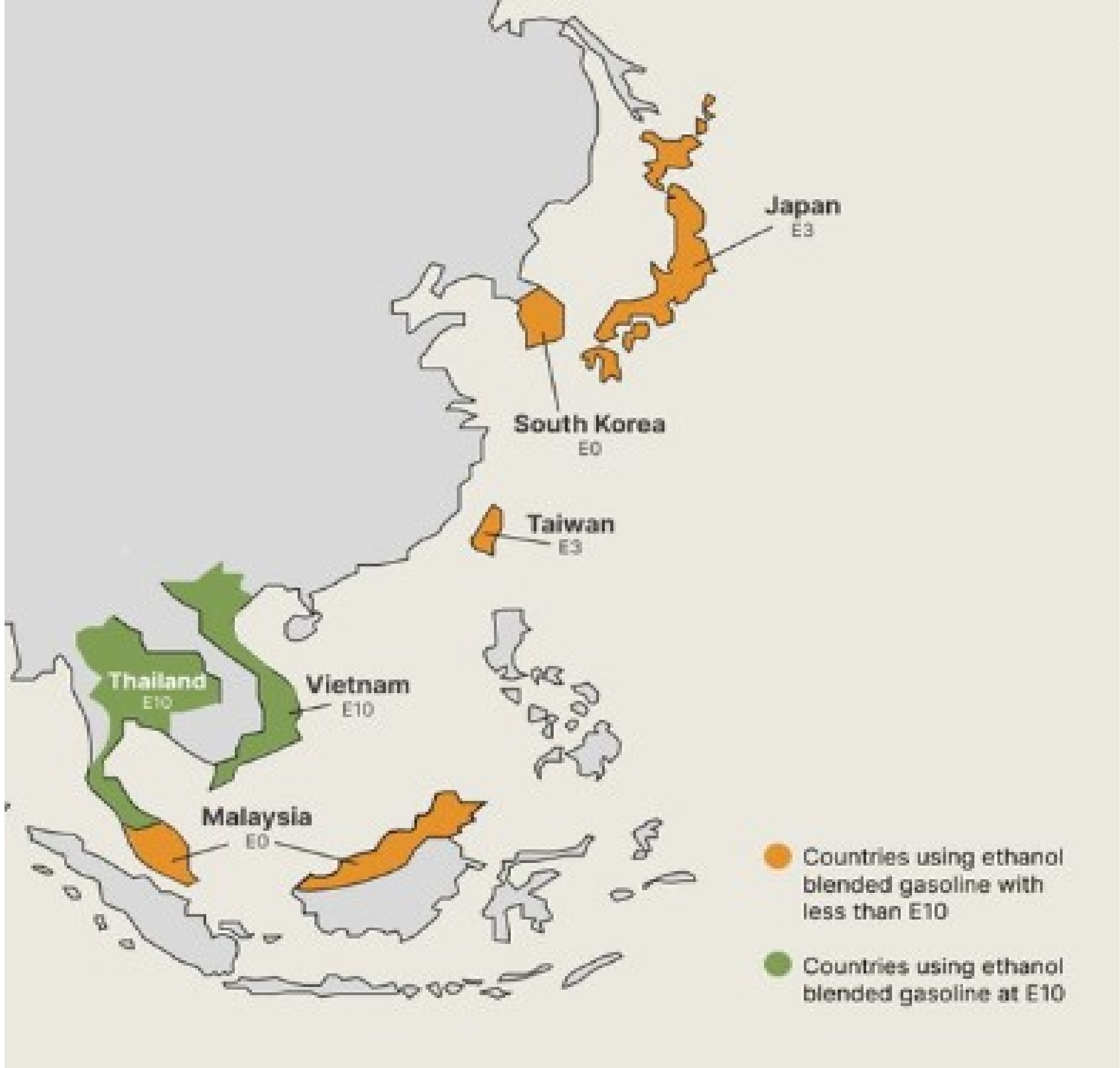
Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

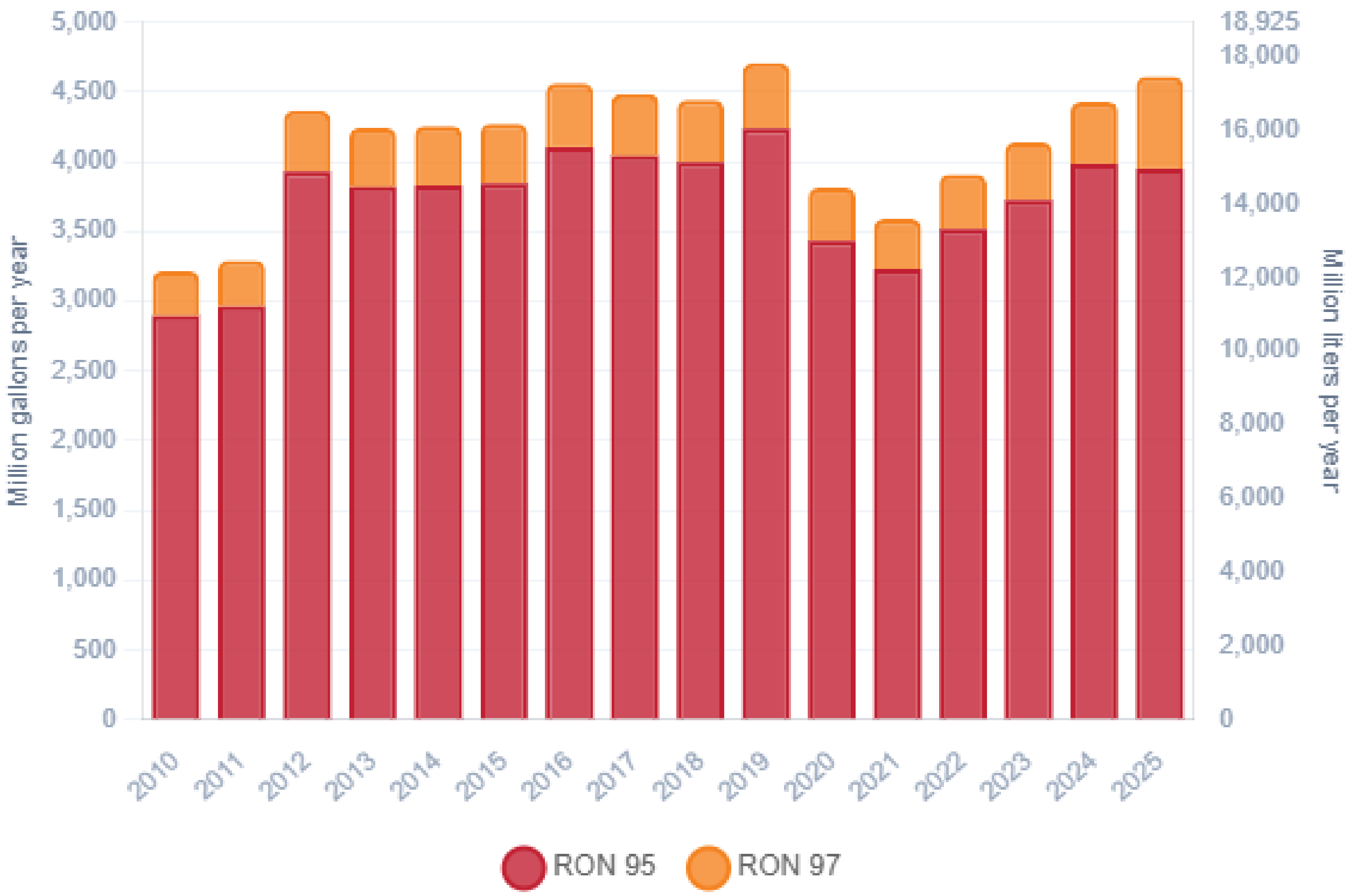
Ethanol Gasoline Blending in Malaysia

In 2025, gasoline consumption in Malaysia was 17,472 million liters (4,616 million gallons) and growth rate was 2.4%. Gasoline production and net imports were 6,864 and 10,608 million liters (1,814 and 2,803 million gallons) correspondingly. Malaysia offers two main grades: RON 95 and RON 97) with an average octane of 95.3 RON.

Government support for biofuels has so far targeted biodiesel. The Biofuel Industry Act of 2007 excluded ethanol as the source of alternative fuels under the National Biofuels Policy of 2006. As there is no policy support for ethanol and there is no production of fuel ethanol and minimal consumption. In 2025, ethanol consumption in Malaysia was 16 million liters (4 million gallons) representing 0.1% of gasoline demand.

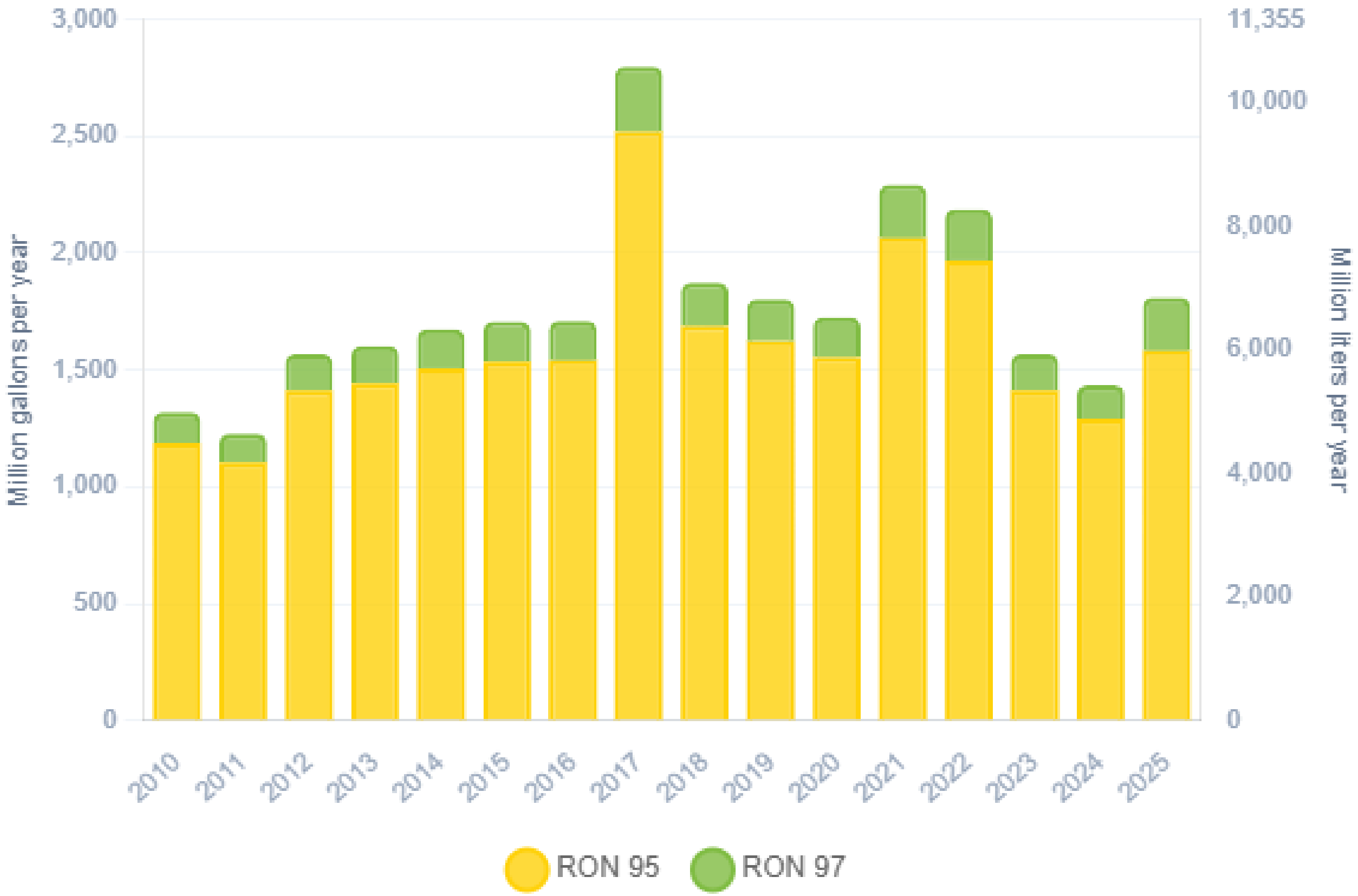
Source: EIA, Suruhanjaya Tenanga, USDA, USGrains

Gasoline Demand in Malaysia



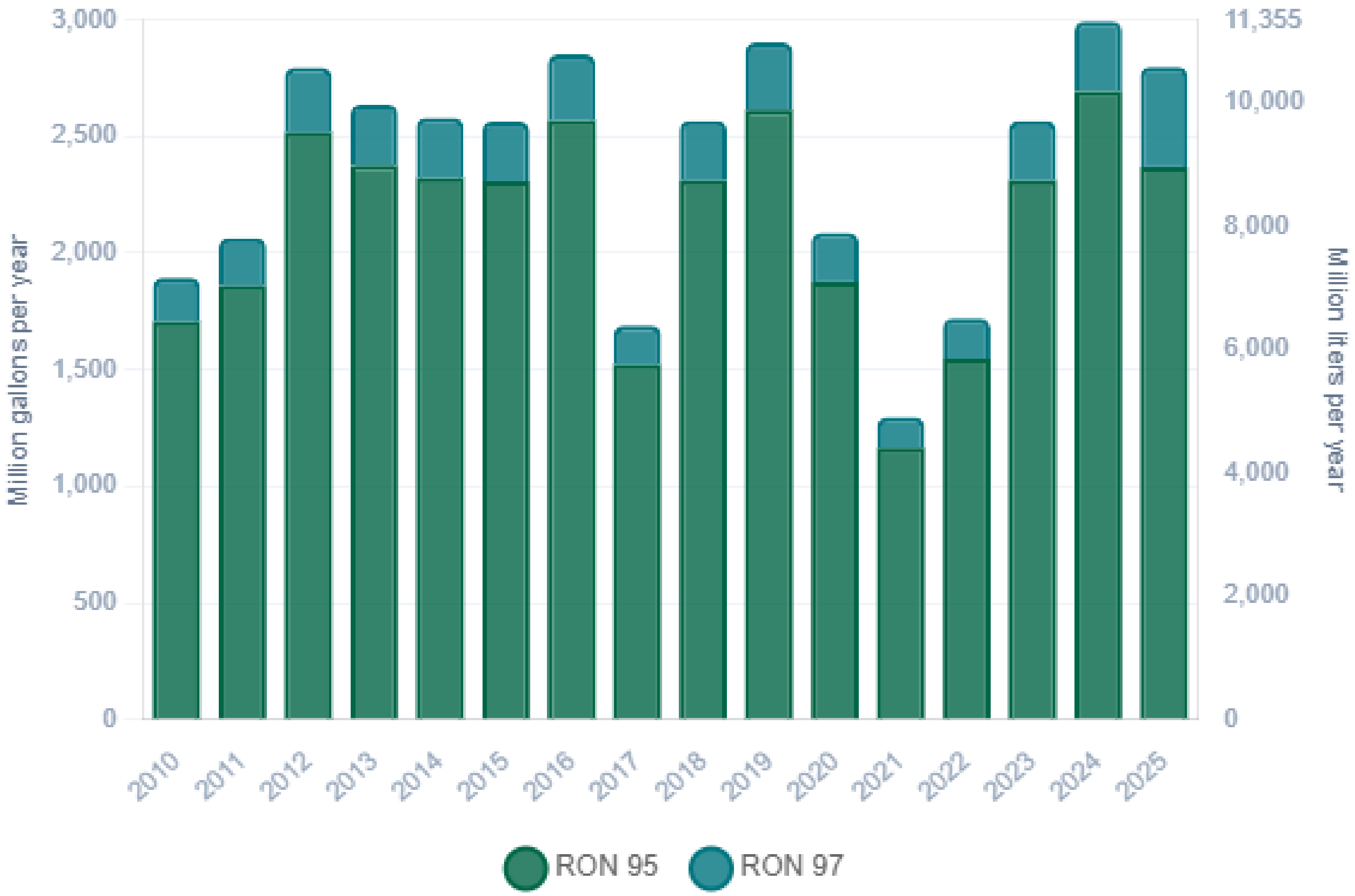
Source: EIA, Suruhanjaya Tenanga

Gasoline Production in Malaysia



Source: EIA, Suruhanjaya Tenanga

Gasoline Net Imports in Malaysia



Source: EIA, Suruhanjaya Tenanga

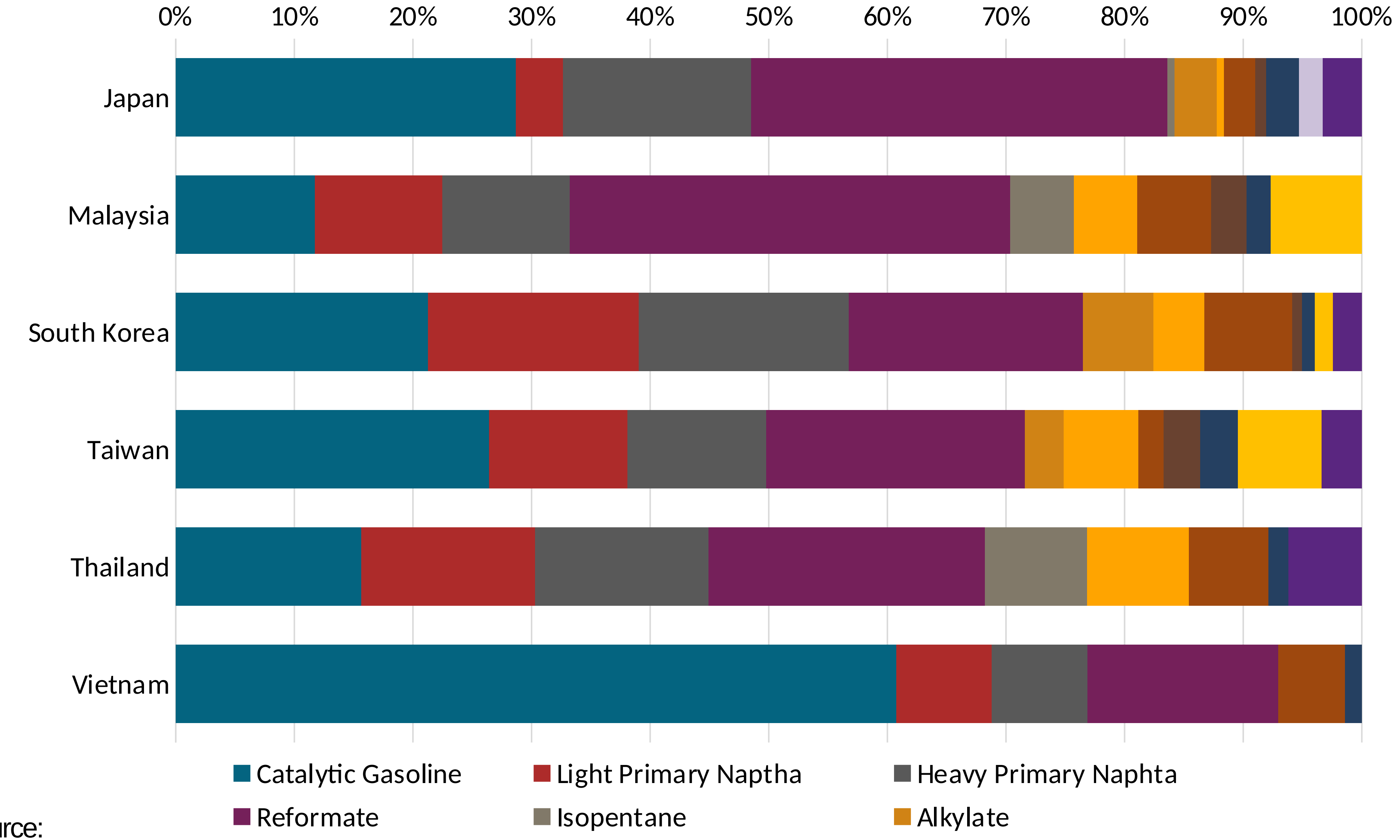
Gasoline Quality in Malaysia

Source: Department of Standards Malaysia, Euro 4 Spec

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

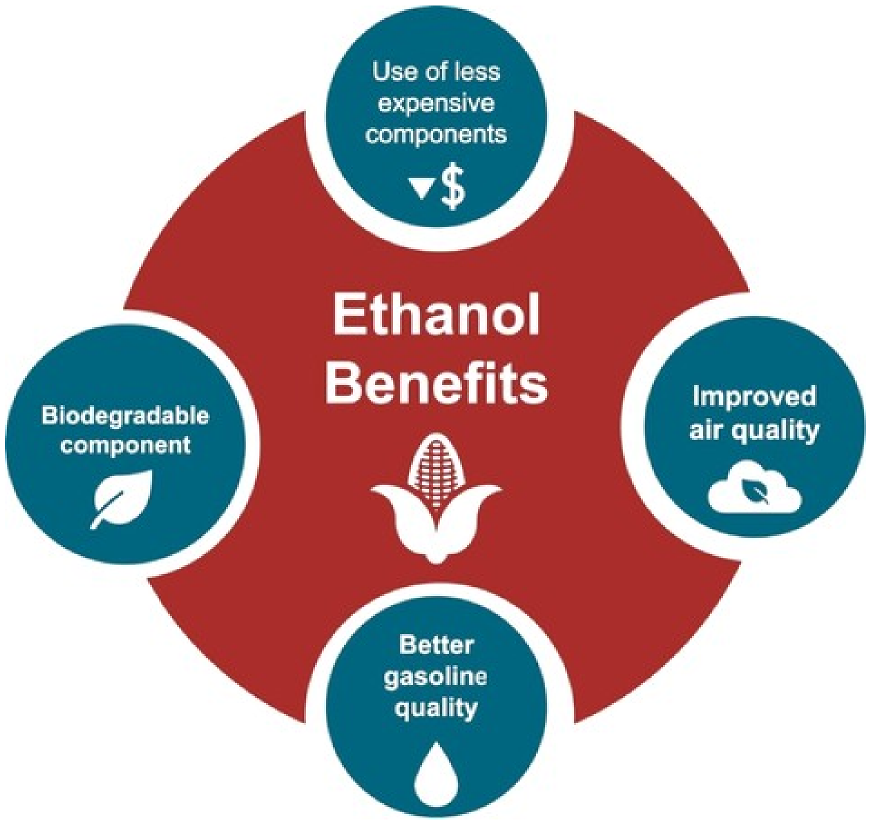
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

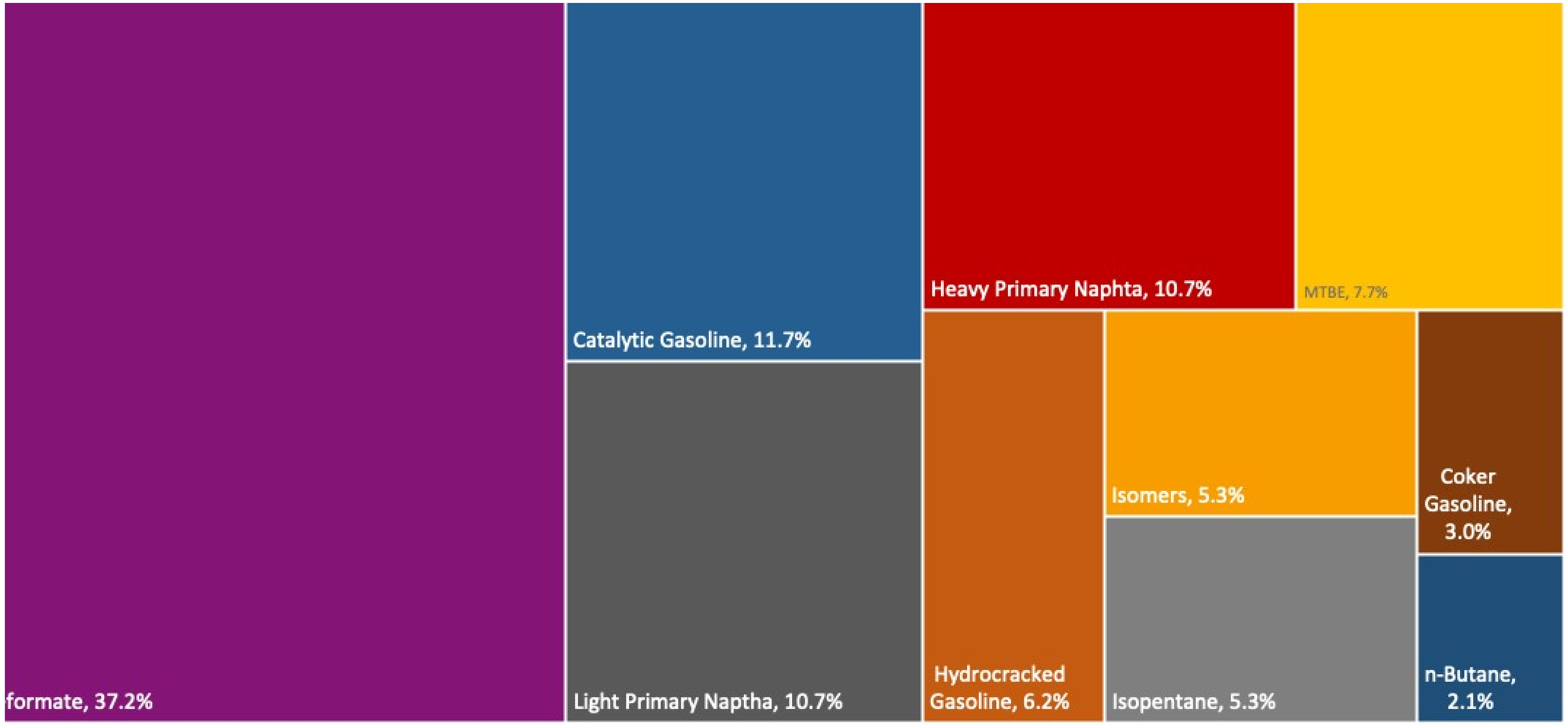
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

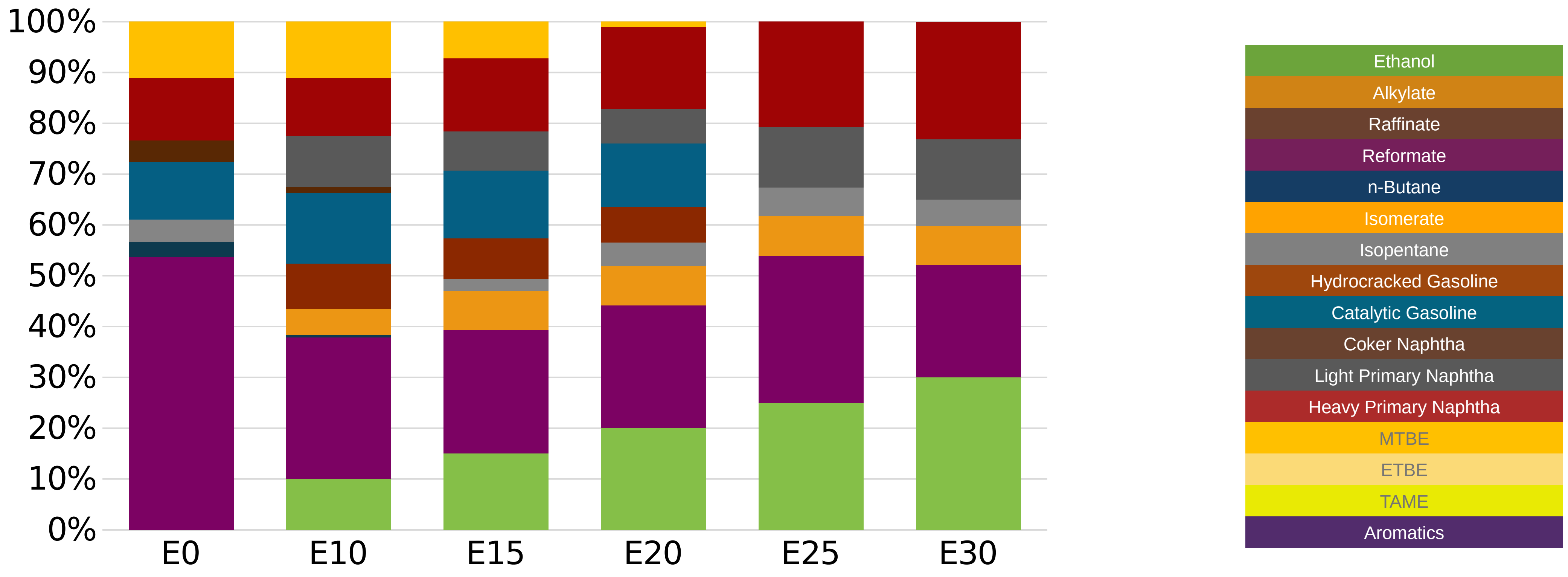


Malaysia Available Blending Components



Source:
Faro90

Malaysia – Gasoline 95 – Constant Octane

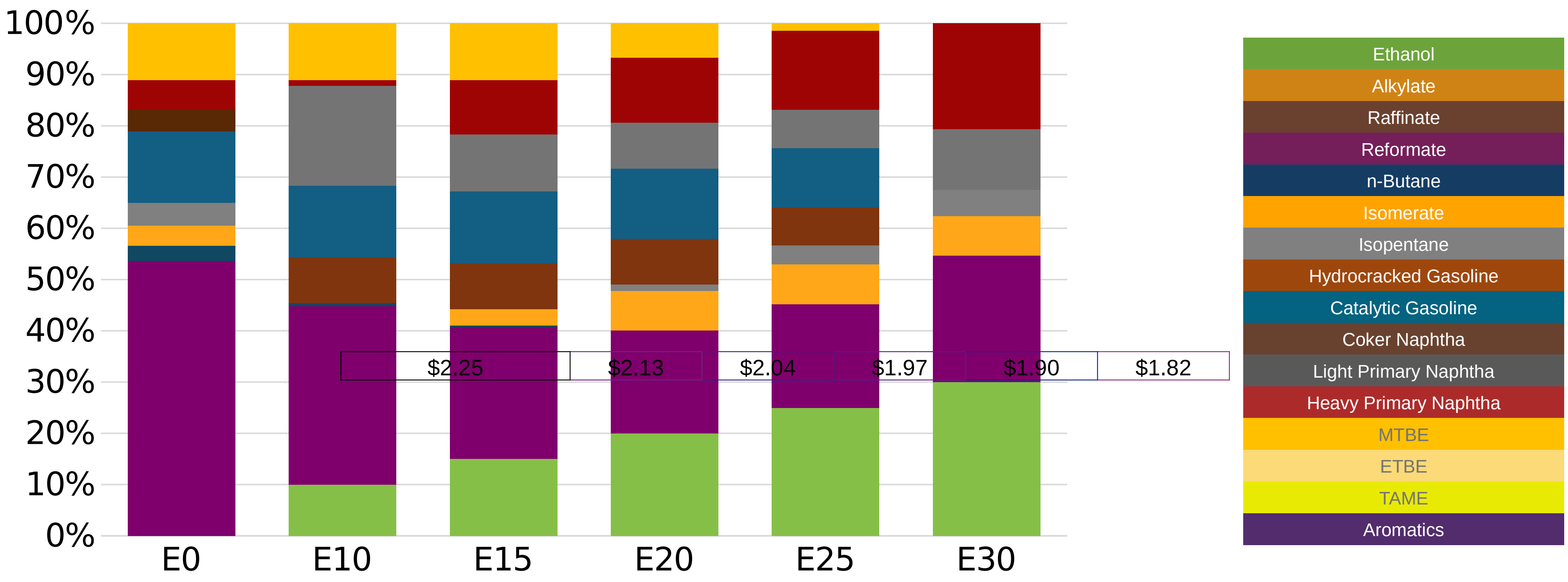


Octane (RON)	95.0	95.0	95.0	95.0	95.6	96.6
Price (US$/gal)	$2.15	$2.02	$1.95	$1.89	$1.81	$1.76

Average CIF 2025 Prices

Source: Faro90

Malaysia - Gasoline 97 - Constant Octane

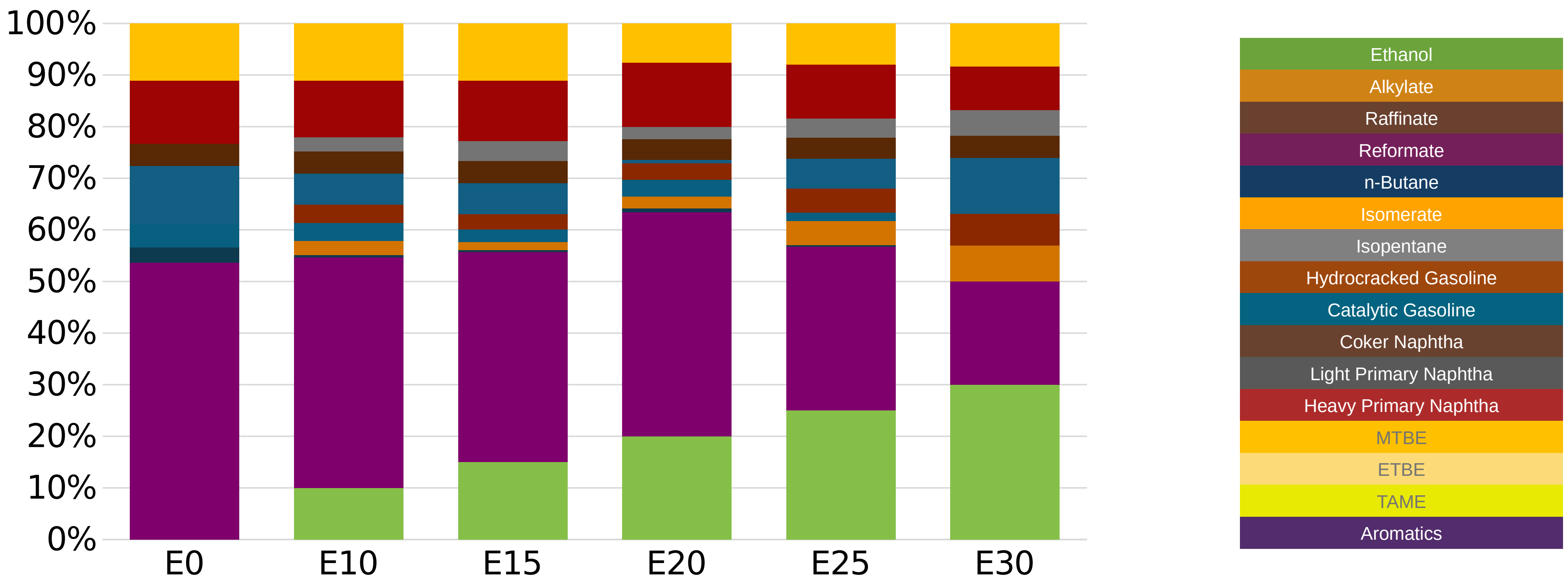


Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.5
Price (US$/gal)	$2.25	$2.13	$2.04	$1.97	$1.90	$1.82

Average CIF 2025 Prices

Source: Faro90

Malaysia – Gasoline 95 – Increased Octane

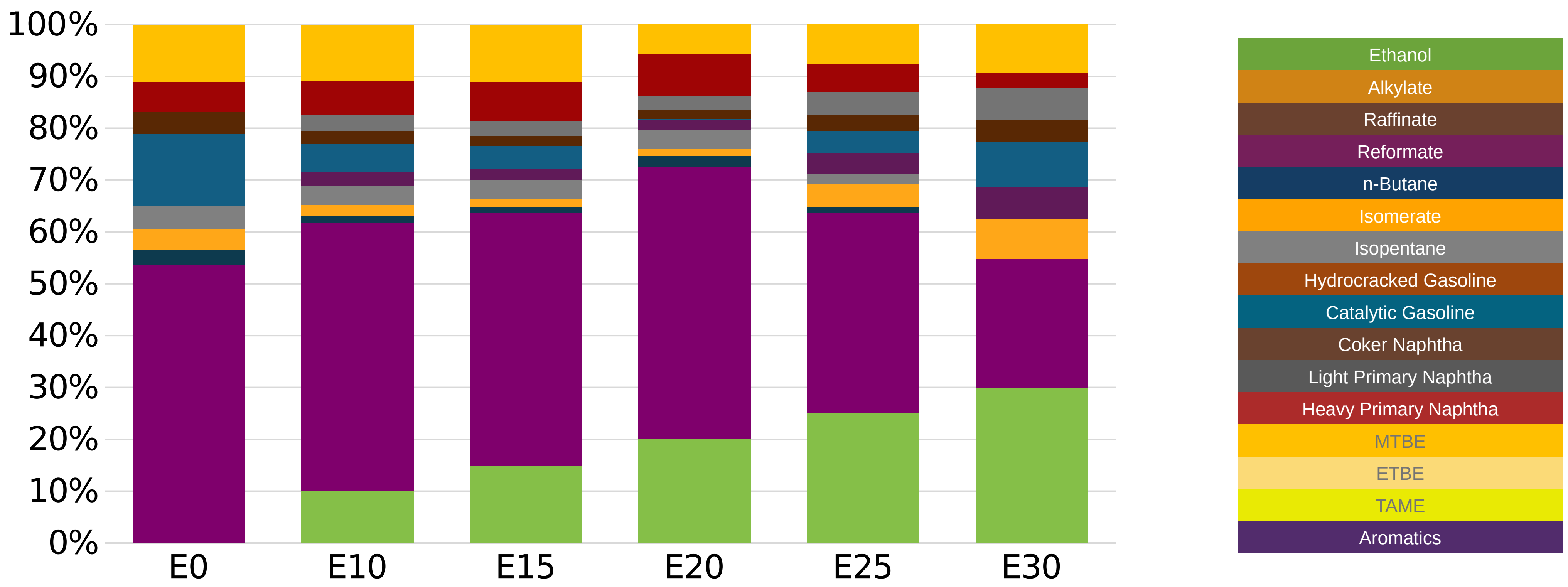


Octane (RON)	95.0	97.8	99.4	100.5	101.2	103.2
Price (US$/gal)	$2.15	$2.15	$2.15	$2.15	$2.15	$2.15

Average CIF 2025 Prices

Source: Faro90

Malaysia – Gasoline 97 – Increased Octane



Octane (RON)	97.0	99.8	101.3	102.1	102.9	104.9
Price (US$/gal)	$2.25	$2.25	$2.25	$2.25	$2.25	$2.25

Average CIF 2025 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

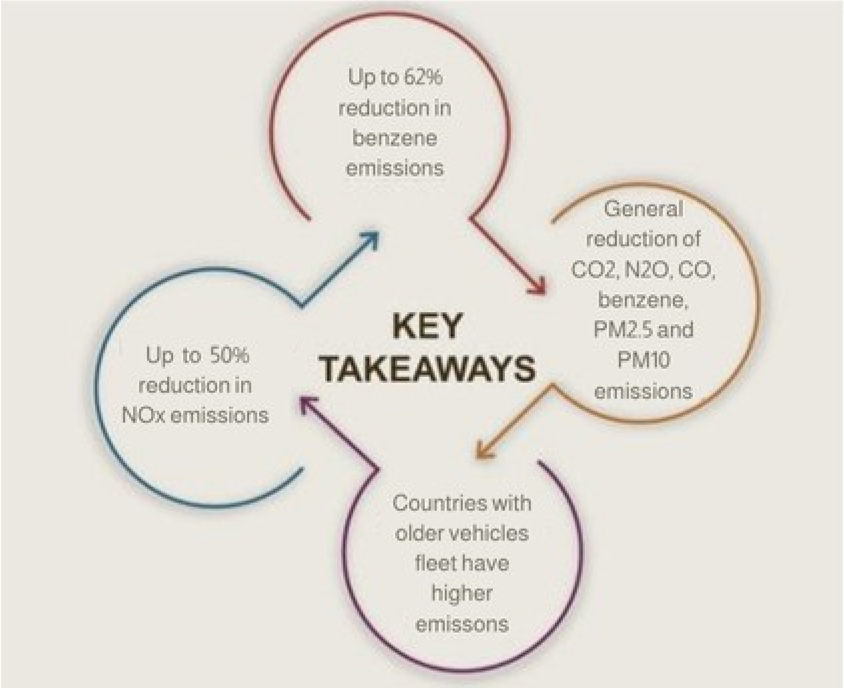
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

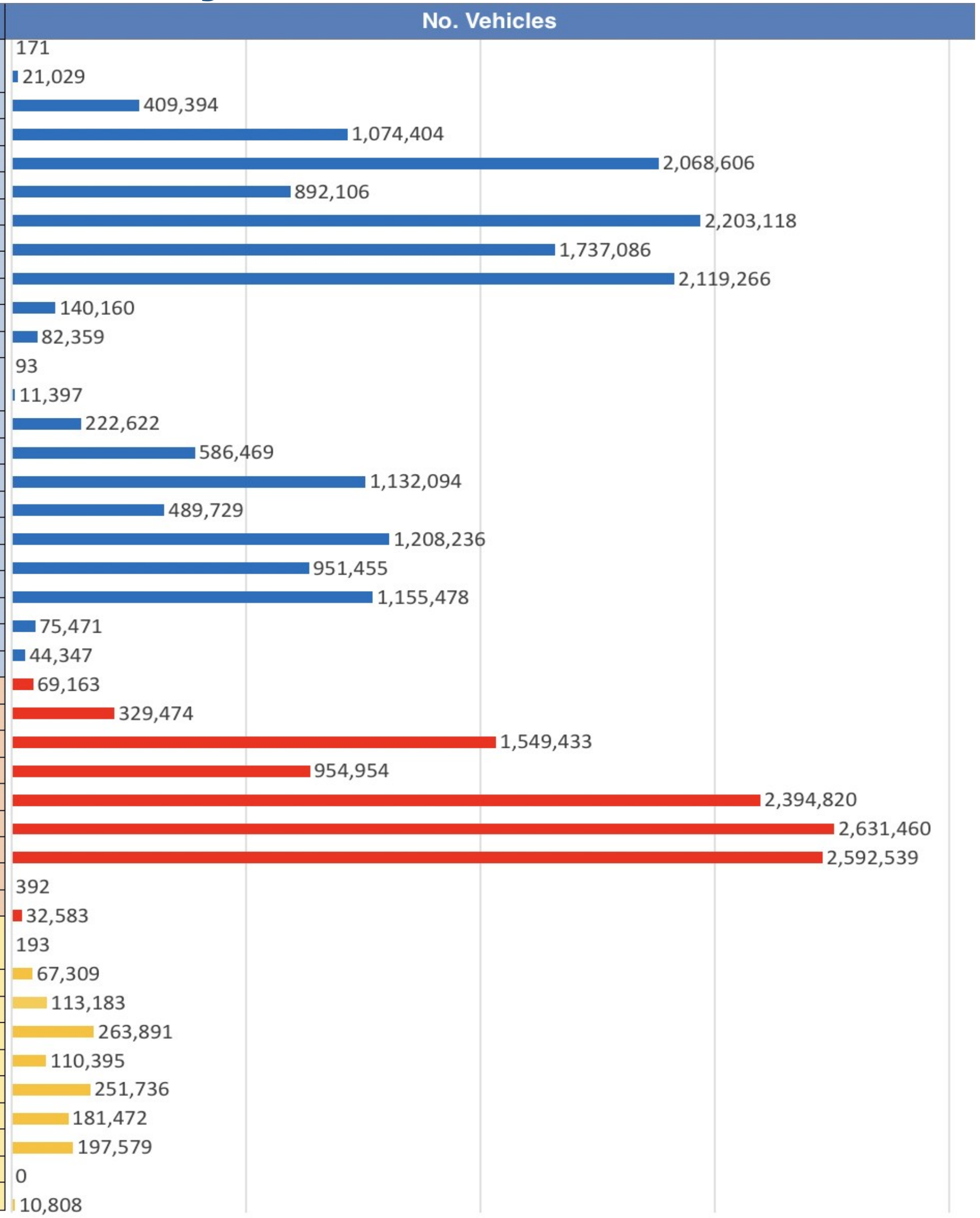
Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - Malaysia

Type	Weight	Engine	Exhaust	Recovery	Age
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years
		Single FI	3-Ways		>8 years
		Multi FI	3-Ways		>8 years
			Euro III		>8 years
			Euro IV		>8 years
			Euro V		>8 years
			Euro 6		>8 years
					4-8 years
					<4 years
			Hybrid		All
		Electric			All
	Medium	Carburator	3-Ways	PCV Tank	>8 years
		Single FI	3-Ways		>8 years
		Multi FI	3-Ways		>8 years
			Euro III		>8 years
			Euro IV		>8 years
			Euro V		>8 years
			Euro 6		>8 years
					4-8 years
					<4 years
			Hybrid		All
		Electric			All
Sml Engines	Light	4-Cycle	None	PCV	>8 years
			Euro III		>8 years
			Euro IV		>8 years
			Euro V		>8 years
			Euro 6		>8 years
					4-8 years
					<4 years
			Hybrid		All
		Electric			All
Truck	Heavy	Carburator	3-Ways	PCV	>8 years
		Fuel-Injection	3-Ways		>8 years
			Euro III		>8 years
			Euro IV		>8 years
			Euro V		>8 years
			Euro 6		>8 years
					4-8 years
					<4 years
			Hybrid		All
		Electric			All

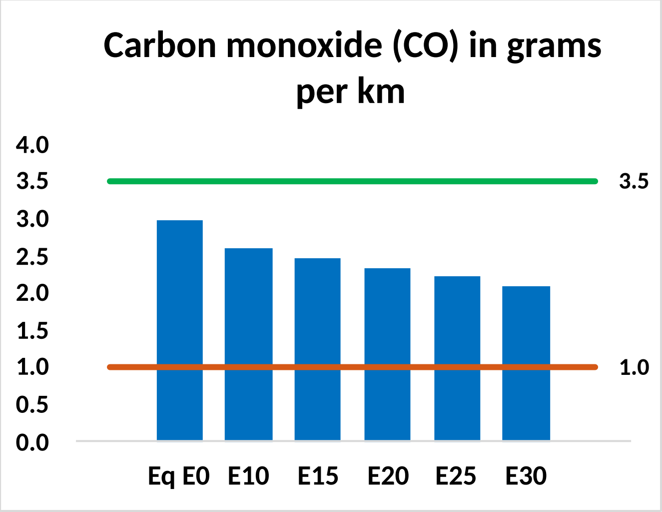


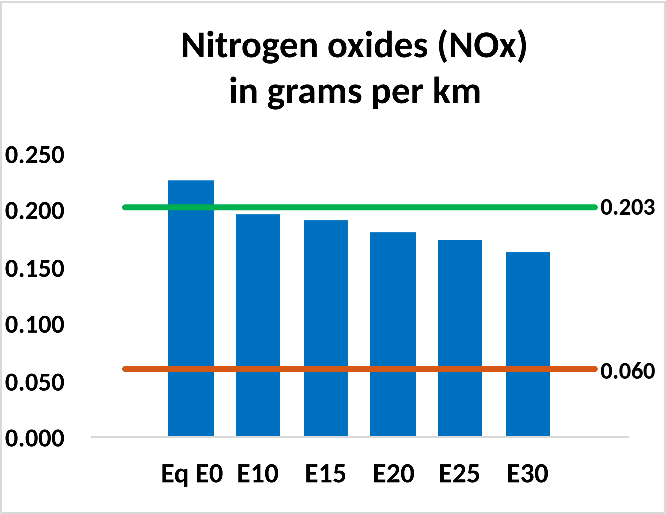
Vehicle Fleet: 28,376,474
Gasoline Vehicle Fleet: 25,554,393

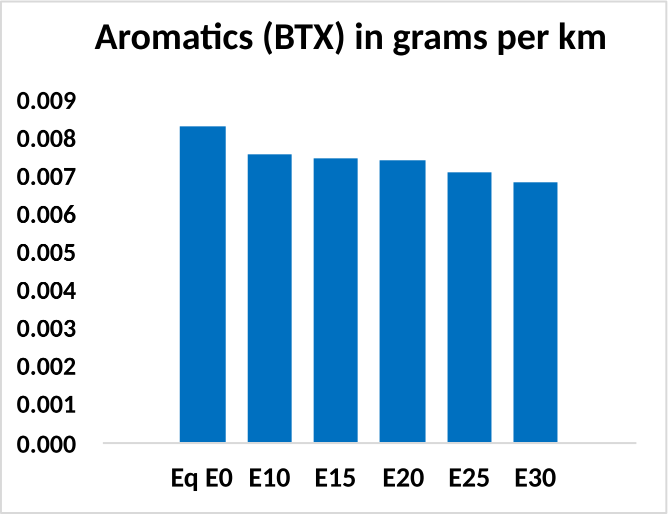
Source: Data Gov Malaysia

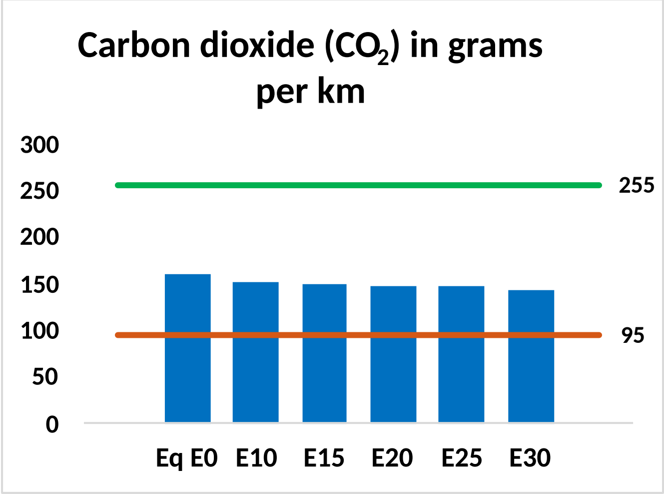
Malaysia – Vehicular Emissions

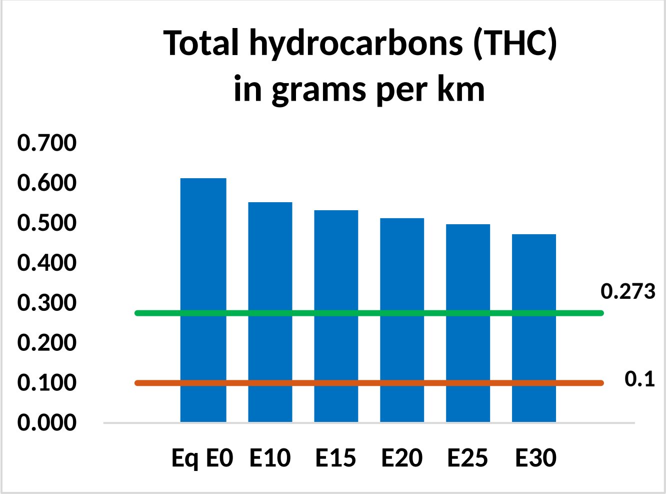
TIER III BIN 125 United States
Euro 6 Standard

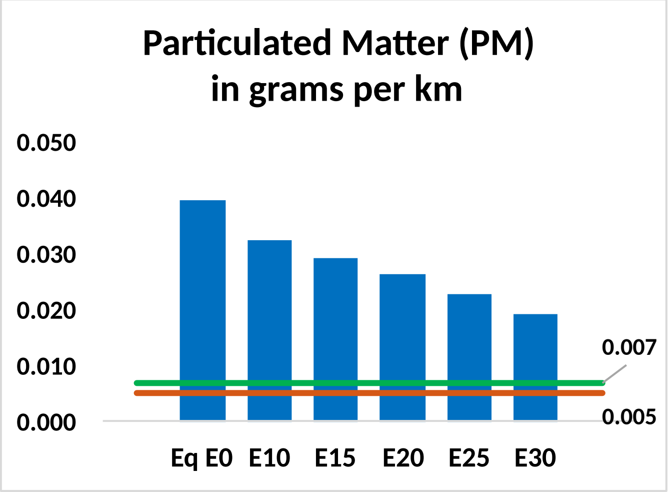












Source: Faro90

Malaysia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,787,897	1,676,608	1,565,319	1,479,227	1,394,939	1,330,835	1,253,790	-12%	-22%
CO2	96,171,362	93,698,384	91,362,794	89,525,921	88,616,964	88,187,848	86,562,292	-5%	-8%
VOC	367,966	349,915	331,865	319,071	307,115	298,200	285,323	-10%	-17%
NOx	136,847	126,355	119,107	115,389	109,428	104,743	99,234	-13%	-20%
SOx	1,106	1,022	939	868	800	727	649	-15%	-28%
NH3	36,424	36,063	35,703	35,872	36,276	36,560	36,795	-2%	0%
N2O	1,264	1,257	1,252	1,258	1,284	1,298	1,313	-1%	2%
CH4	88,011	88,011	88,011	89,331	91,267	92,940	94,524	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,124	2,035	1,946	1,888	1,836	1,798	1,741	-8%	-14%
Acetaldehyde	6,537	7,503	11,009	16,765	22,840	26,100	30,839	68%	249%
Formaldehyde	26,261	25,532	29,763	34,948	36,613	40,504	44,093	13%	39%
Benzene	5,017	4,822	4,569	4,496	4,459	4,264	4,127	-9%	-11%
PM2.5	6,229	5,663	5,096	4,598	4,110	3,588	3,041	-18%	-34%
PM10	17,715	16,105	14,494	13,077	11,690	10,204	8,648	-18%	-34%

Source: Faro90

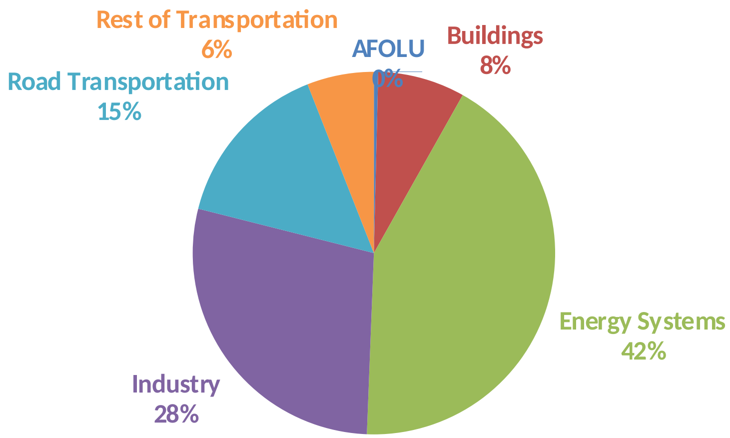
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

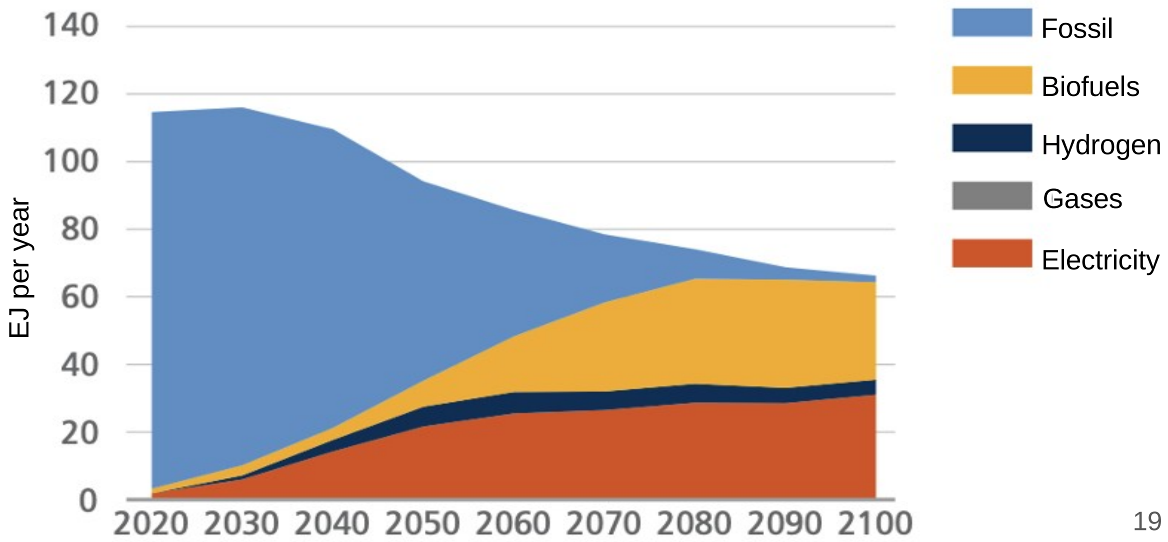
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

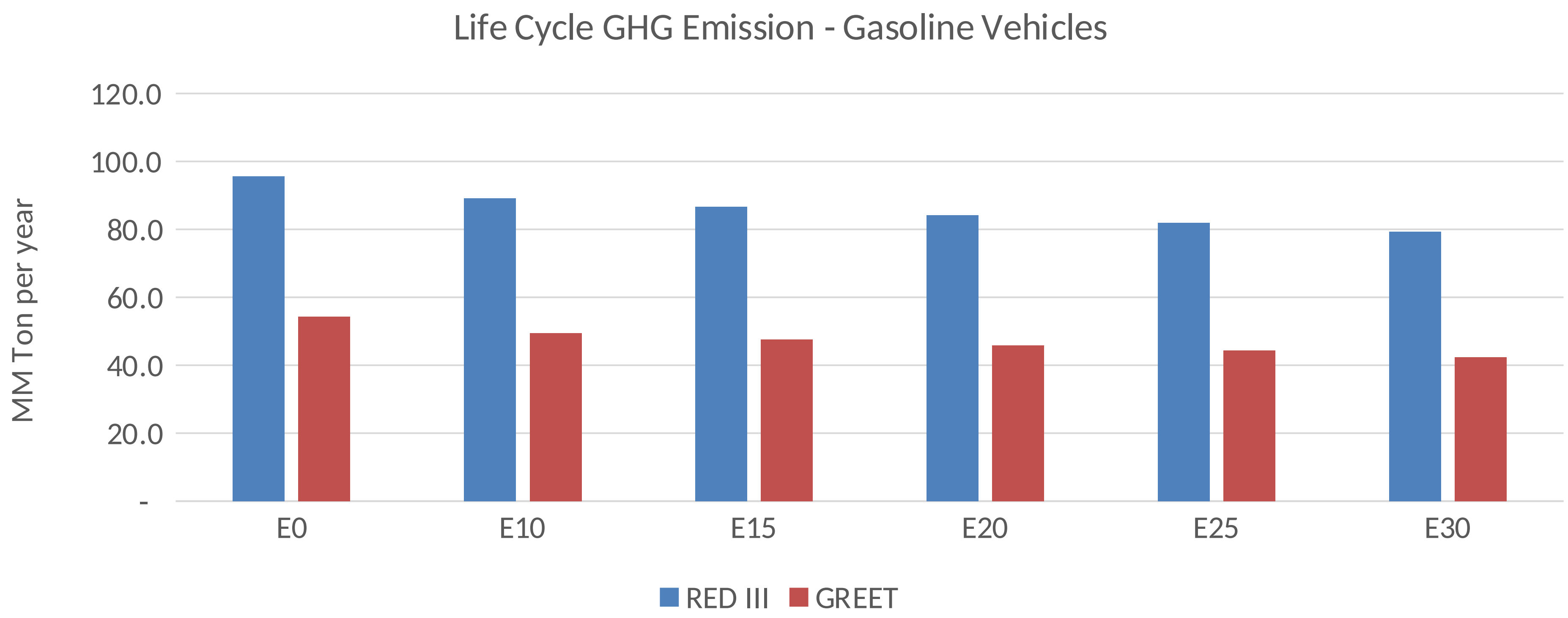
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Malaysia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	407.7
Target @ 2035 (MMT/year)	285.4
Delta	122.3

Source: Greet, RED II/III, climateactiontracker.org, F90